

This problem set has 4 problems in total.

****Depending on your computer some models may take hours to fit → don't wait until the last minute to complete this problem set****

P1. Sometimes the prior and the data (through the likelihood) are in conflict because they concentrate around different regions of parameter space. What happens in these cases depends a lot upon the shape of the tails of the distributions (e.g., <https://doi.org/10.1111/j.2517-6161.1979.tb01090.x>). Likewise, the tails of distributions strongly influence whether outliers are shrunk toward the mean or not.

Consider four different models to fit to one single observation at $y = 0$. The models differ only in the distributions assigned to the likelihood and prior. Here are the four models:

Model NN: $y \sim \text{Normal}(\mu, 1)$ $\mu \sim \text{Normal}(10, 1)$

Model TN: $y \sim \text{Student}(2, \mu, 1)$ $\mu \sim \text{Normal}(10, 1)$

Model NT: $y \sim \text{Normal}(\mu, 1)$ $\mu \sim \text{Student}(2, 10, 1)$

Model TT: $y \sim \text{Student}(2, \mu, 1)$ $\mu \sim \text{Student}(2, 10, 1)$

Estimate the posterior distributions for these models and compare them. Can you explain the results, using the properties of the distributions? An example of how to estimate these models with `ulam` is given below for model NN.

```
mnn <- ulam(  
  alist(  
    y ~ dnorm(mu,1),  
    mu ~ dnorm(10,1)  
  ), data=list(y=0) , chains=4 )
```

P2. This task uses the trolley problem data we discussed in class (Trolley.csv). Define and fit ordinal regression models for these data that either have no varying effects or include varying effects that cluster intercepts on individual participants. Use a non-centered parameterization for the varying effects model.

In the data frame, the participants are indicated by the unique values in the “id” variable. Note that you may need to create a 1:N index variable for these ids rather than passing the string values of id directly into the model. This depends on the analysis package you use; some will do this conversion for you (ulam will not). Include action, intention, contact, intention*action, and intention*contact as ordinary terms in the regression model (i.e., without varying effects). Note that action and contact never occur in the same trial so there is no interaction term for those treatments.

Compare the varying intercepts model and a model that ignores individuals (i.e., no varying intercepts), using both WAIC and posterior estimates of the mean and SD for the treatment effects. What is the impact of individual variation in these data?

To avoid confusion about the specification of the core model, here is the ulam code for the version without any varying effects.

```
d <- Trolley
dat <- list(
  R = d$response,
  A = d$action,
  I = d$intention,
  C = d$contact,
  Sid = coerce_index(d$story),
  id = coerce_index(d$id) )

m <- ulam(
  alist(
    R ~ dordlogit( phi , cutpoints ),
    phi <- bA*A + bC*C + BI*I ,
    BI <- bI + bIA*A + bIC*C ,
    c(bA,bI,bC,bIA,bIC) ~ dnorm( 0 , 0.5 ),
    cutpoints ~ dnorm( 0 , 1.5 )
  ) , data=dat , chains=4 , cores=4 , log_lik=TRUE )
```

Note that setting log_lik = TRUE saves the log likelihoods for use with WAIC or PSIS-LOO model comparisons.

P3. The Trolley data are also clustered by story, which indicates a unique narrative for each vignette. Define and fit a cross-classified varying intercepts ordinal regression model with both “id” and “story” as varying effects (non-centered parameterization). Use the same ordinary terms as in the previous problem. Compare this model to the previous models. What do you infer about the impact of different stories on responses? Does the rank order of the treatment-type effect sizes remain constant across models with versus without varying effects for stories? If they change, why might this be?

P4. This problem uses the data in `Primates301.csv` to examine the relationship between body mass and brain size. Here we will consider missing values—this data set has many missing entries. There are body mass measurements for 238 species and 56 missing brain values among those 238 cases. Use the subset of the data with measurements for body mass (i.e., 238 species). There are 2 species for which we somehow have brain but not body mass; we will drop those two for simplicity.

First, consider whether there is a pattern to the missing values. Does it look like missing brain values are associated with particular values of body mass? Draw a DAG that represents how missingness works in these data. Which type (MCAR, MAR, MNAR) is this?

Second, compare a model (see below) fit to only the complete cases versus one that imputes missing values for brain size. It might help to initialize the 56 imputed variables to a valid value. For `ulam`, you could add this line to your model fitting code: `start=list(B_impute=rep(0.5,56))`. This just helps the chain get started.

Does anything change between the two models? Why or why not? Hint: Consider the density of data in the ranges where there are missing values. You might want to plot the imputed brain sizes together with the observed values. If you fit with `ulam`, the estimated brain sizes are the `B_impute[1] ... B_impute[56]` coefficients from the `precis` output.

To avoid any confusion regarding the model specification, here is the model code for the analysis using only complete cases.

```
d <- Primates301
cc <- complete.cases( d$brain, d$body)
B <- d$brain[cc]
M <- d$body[cc]
B <- B/ max(B)
M <- M/ max(M)

dat_list <- list( B = B, M = M)
m <- ulam(
  alist(
    B ~ dlnorm( mu, sigma),
    mu <- a + b*log(M),
    a ~ normal(0,1),
    b ~ normal(0,1),
    sigma ~ exponential(1)
  ),
  data=dat_list)
```

*Be careful when scaling the B vector that includes missing values – you need to compute the max ignoring NAs because otherwise the whole vector becomes NA and you can't impute an entirely missing outcome vector!